

# Design and Parasitic-Aware Post-Layout Realization of a 6.5 GHz Three-Stage CMOS Ring Oscillator in a 65 nm Technology

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**Abstract**—This paper presents the complete custom-design methodology used to realize and verify a 6.5 GHz three-stage CMOS ring oscillator in a 65 nm CMOS technology operating from a 1 V supply. The emphasis of the work is on the full physical-design and verification flow that yields a credible, parasitic-aware oscillation frequency, rather than on idealized schematic estimates. The oscillator is built from three cascaded static CMOS inverter stages connected in a feedback loop. The methodology covers manual layout, functional design-rule checking (DRC), layout-versus-schematic (LVS) verification, and parasitic resistance–capacitance (RC) extraction in a detailed standard parasitic format (DSPF). Simulation of the RC-extracted netlist yields an oscillation frequency of 6.50 GHz with a power consumption of 286.3  $\mu$ W from a 1 V supply, while occupying a core area of only 73.96  $\mu$ m<sup>2</sup>. A periodic steady-state and periodic-noise analysis of the same RC-extracted netlist gives a phase noise of  $-77.6$  dBc/Hz at a 1 MHz offset, corresponding to a figure of merit (FoM) of  $-159.3$  dBc/Hz. The flow demonstrates that a parasitic-aware methodology is essential for an accurate high-frequency ring-oscillator characterization, since interconnect parasitics substantially lower the oscillation frequency relative to pre-layout estimates.

**Index Terms**—Ring oscillator, CMOS, 65 nm, physical design, DRC, LVS, parasitic extraction, post-layout simulation, phase noise, DSPF.

## I. INTRODUCTION

Ring oscillators are widely used in modern integrated circuits as the core of phase-locked loops (PLLs), clock and data recovery circuits, and on-chip clock generators [1]. Compared with *LC* oscillators, they offer a wide tuning range, occupy a much smaller die area, and can be implemented in any standard CMOS process without dedicated passive components [10], [11]. Among the possible topologies, the three-stage ring has been demonstrated successfully in many works [10], which makes it an attractive and well-understood configuration.

A common shortcoming in reported designs is that performance figures are quoted from schematic-level simulation, which neglects the parasitics introduced by the physical layout. In a deep-submicron technology operating at multi-gigahertz

TABLE I  
TRANSISTOR DIMENSIONS

| Device           | Type | <i>W</i>  | <i>L</i> |
|------------------|------|-----------|----------|
| NMOS (3 devices) | nch  | 2 $\mu$ m | 60 nm    |
| PMOS (3 devices) | pch  | 4 $\mu$ m | 60 nm    |

frequencies, interconnect resistance and capacitance significantly affect the node delays and therefore the oscillation frequency. A meaningful characterization must therefore be performed on a layout that has been verified (DRC and LVS) and whose parasitics have been extracted.

This paper focuses on the *methodology* required to obtain such a parasitic-aware result. The main contributions are: (i) a complete and reproducible custom-design flow—layout, functional DRC, LVS, and RC extraction—for a three-stage 65 nm CMOS ring oscillator; (ii) a post-layout, parasitic-aware characterization yielding 6.50 GHz at 286.3  $\mu$ W in a 73.96  $\mu$ m<sup>2</sup> core, including a phase-noise and figure-of-merit characterization extracted from the same RC netlist; and (iii) a discussion of how the parasitics extracted from the layout shape the final oscillation frequency.

The remainder of the paper is organized as follows. Section II describes the circuit architecture. Section III details the design and verification methodology. Section IV presents the post-layout simulation results. Section V concludes.

## II. CIRCUIT ARCHITECTURE

The oscillator is a three-stage inverting ring. Each stage is a static CMOS inverter consisting of one PMOS pull-up device and one NMOS pull-down device; the output of the third stage is fed back to the input of the first stage to close the oscillating loop. The transistor dimensions are summarized in Table I.

To first order, the period of an *N*-stage ring oscillator is set by the propagation delay  $t_d$  of a single stage,

$$f_{osc} = \frac{1}{2N t_d}, \quad t_d \approx \frac{C_{node} \Delta V}{I_{drive}}, \quad (1)$$

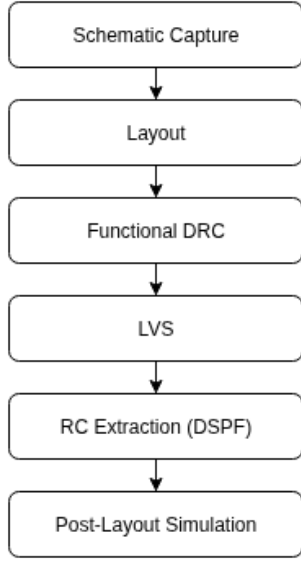


Fig. 1. Custom physical-design and parasitic-aware verification flow used in this work.

where  $N = 3$ ,  $C_{node}$  is the total capacitance loading a node,  $\Delta V$  is the switching voltage, and  $I_{drive}$  is the available charging/discharging current. Equation (1) shows that the oscillation frequency is inversely proportional to the node capacitance; consequently, the interconnect parasitics added by the layout directly reduce  $f_{osc}$ . This dependence motivates the parasitic-aware methodology described next.

### III. DESIGN AND VERIFICATION METHODOLOGY

The custom flow used in this work is summarized in Fig. 1. After schematic capture, the circuit was laid out manually, checked against the functional design rules, verified for schematic equivalence (LVS), and finally subjected to parasitic RC extraction before post-layout simulation.

#### A. Layout

The ring oscillator was laid out in the target 65 nm CMOS technology. As shown in Fig. 2, the three PMOS devices are placed in a common  $n$ -well along the upper row and share the  $V_{DD}$  rail, while the three NMOS devices occupy the lower row and share the ground rail. The three inter-stage nodes ( $V_1$ ,  $V_2$ , and the output) are routed in the lower metal layers to keep the interconnect short. The resulting core occupies an  $8.6 \mu\text{m} \times 8.6 \mu\text{m}$  region, i.e. a core area of  $73.96 \mu\text{m}^2$ .

#### B. Functional Design-Rule Check (DRC)

DRC was performed to ensure that the drawn geometry complies with the manufacturing rules of the technology. All *functional* rules, such as minimum spacing and width, were resolved; for example, a metal-1 spacing violation (minimum  $0.09 \mu\text{m}$ ) was corrected by adjusting the offending geometry. The remaining DRC items are chip-level seal-ring rules and foundry density rules (metal, poly, and diffusion density), together with informational reminders. These pertain to full-chip integration and manufacturing uniformity rather than to

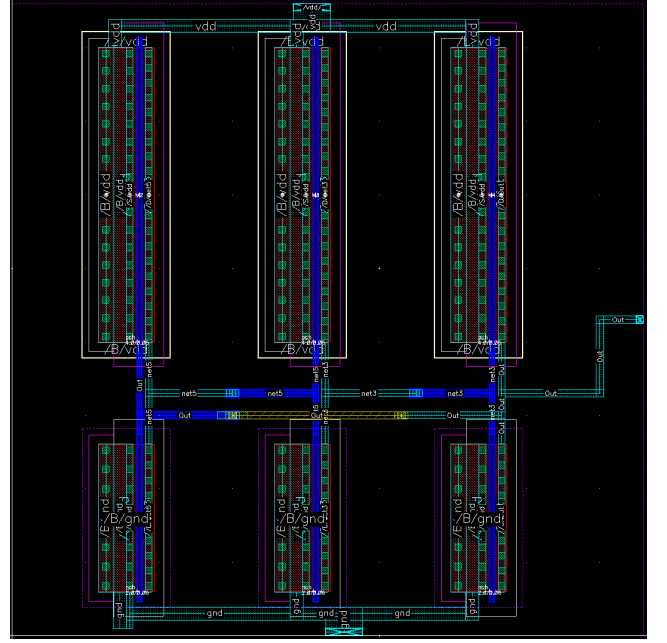


Fig. 2. Layout of the three-stage CMOS ring oscillator core ( $8.6 \mu\text{m} \times 8.6 \mu\text{m}$ ). The upper row contains the three PMOS devices and the lower row the three NMOS devices; the inter-stage nets and the output are routed between them.

TABLE II  
EXTRACTED PER-NET PARASITIC CAPACITANCE

| Net      | Parasitic capacitance (fF) |
|----------|----------------------------|
| Out      | 4.07                       |
| $V_{DD}$ | 3.16                       |
| gnd      | 3.01                       |
| net3     | 3.54                       |
| net5     | 3.52                       |

the functional correctness of the oscillator core, and therefore do not affect the extracted netlist or the simulated behavior.

#### C. Layout-versus-Schematic (LVS)

LVS was carried out to confirm that the layout is electrically equivalent to the schematic. A clean LVS match was obtained, with no shorts, opens, or unmatched devices. The LVS-clean result is the essential prerequisite for parasitic extraction and post-layout simulation, since it guarantees that the extracted netlist faithfully represents the intended topology.

#### D. Parasitic RC Extraction

A detailed parasitic extraction was performed on the verified layout, producing a transistor-level netlist annotated with interconnect resistances and capacitances in the detailed standard parasitic format (DSPF). The extraction captures the per-net parasitic capacitances and the metal/poly interconnect resistances. The extracted per-net capacitances of the key nodes are listed in Table II; these values quantify the additional loading that the layout imposes on each oscillating node.

The RC-extracted netlist was then simulated under a 1 V supply using a transient analysis. To guarantee start-up of the

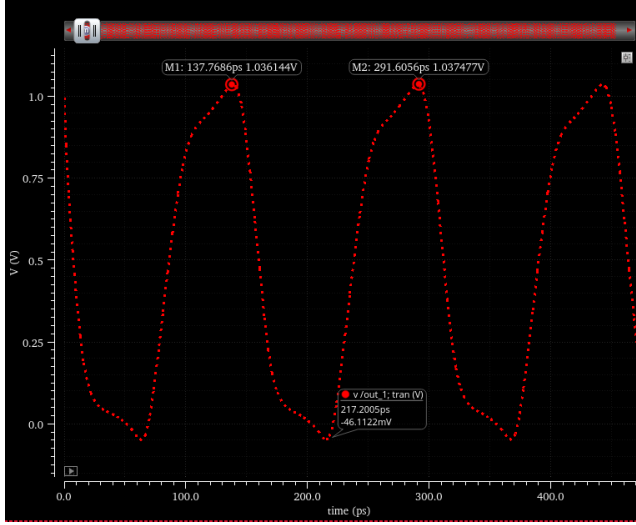


Fig. 3. Post-layout (RC-extracted) transient output waveform. The two markers at 137.77 ps and 291.61 ps yield a period of 153.84 ps, i.e. an oscillation frequency of 6.50 GHz.

oscillation, an initial condition was imposed on the output node.

#### IV. POST-LAYOUT SIMULATION RESULTS

Fig. 3 shows the post-layout transient output waveform obtained from the RC-extracted netlist. The oscillation is stable and spans approximately the full supply range. Two adjacent peaks were measured at  $t_1 = 137.77$  ps and  $t_2 = 291.61$  ps, giving a period

$$T = t_2 - t_1 = 153.84 \text{ ps}, \quad (2)$$

and an oscillation frequency

$$f_{osc} = \frac{1}{T} \approx 6.50 \text{ GHz}. \quad (3)$$

The average supply current was obtained from the transient supply current,  $|I_{avg}| = 286.3 \text{ } \mu\text{A}$ , giving a power consumption of

$$P = V_{DD} |I_{avg}| = 1 \text{ V} \times 286.3 \text{ } \mu\text{A} = 286.3 \text{ } \mu\text{W}. \quad (4)$$

The output swing extends over approximately the full supply range (about  $-0.05$  V to  $1.04$  V), and the per-stage delay implied by (1) is  $t_d = T/(2N) = 25.6$  ps. The complete set of measured performance metrics is summarized in Table III.

##### A. Phase Noise and Figure of Merit

To characterize the spectral purity of the oscillator, a periodic steady-state (PSS) analysis followed by a periodic-noise (pnoise) analysis was performed directly on the RC-extracted netlist. The resulting single-sideband phase-noise characteristic is shown in Fig. 4. The curve exhibits the expected  $1/f^2$  slope across the offset range, confirming a physically consistent result. At a 1 MHz offset from the 6.5 GHz carrier, the phase noise is  $-77.6$  dBc/Hz, decreasing to  $-124.3$  dBc/Hz at a 100 MHz offset.

TABLE III  
SUMMARY OF POST-LAYOUT (RC) PERFORMANCE

| Parameter               | Value                 |
|-------------------------|-----------------------|
| Technology              | 65 nm CMOS            |
| Supply voltage $V_{DD}$ | 1.0 V                 |
| Topology                | 3-stage CMOS ring     |
| Oscillation frequency   | 6.50 GHz              |
| Period                  | 153.84 ps             |
| Stage delay             | 25.6 ps               |
| Power consumption       | 286.3 $\mu\text{W}$   |
| Output swing            | $\sim 1.0$ V          |
| Phase noise @ 1 MHz     | $-77.6$ dBc/Hz        |
| FoM                     | $-159.3$ dBc/Hz       |
| Core area               | 73.96 $\mu\text{m}^2$ |

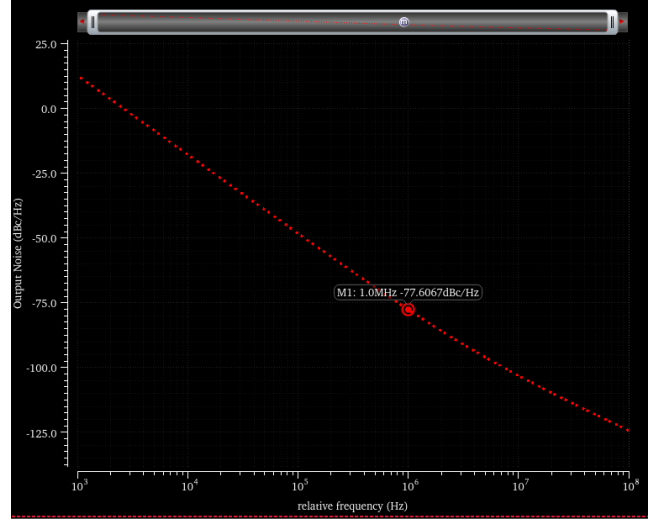


Fig. 4. Post-layout (RC-extracted) single-sideband phase noise versus offset frequency. The marker indicates  $-77.6$  dBc/Hz at a 1 MHz offset.

A widely used figure of merit (FoM) for oscillators normalizes the phase noise to the oscillation frequency and the power consumption [10]:

$$\text{FoM} = \mathcal{L}(\Delta f) - 20 \log_{10} \left( \frac{f_{osc}}{\Delta f} \right) + 10 \log_{10} \left( \frac{P_{dc}}{1 \text{ mW}} \right), \quad (5)$$

where  $\mathcal{L}(\Delta f)$  is the phase noise at offset  $\Delta f$ . Using  $\mathcal{L}(1 \text{ MHz}) = -77.6$  dBc/Hz,  $f_{osc} = 6.5$  GHz,  $\Delta f = 1$  MHz, and  $P_{dc} = 286.3 \text{ } \mu\text{W}$ , the evaluation of (5) gives

$$\text{FoM} = -77.6 - 76.26 - 5.43 \approx -159.3 \text{ dBc/Hz}. \quad (6)$$

This value is competitive with previously reported CMOS ring oscillators [10], indicating that the compact, parasitic-aware design achieves a favorable balance between oscillation frequency, power consumption, and spectral purity.

##### B. Discussion

The post-layout frequency of 6.50 GHz reflects the additional capacitive and resistive loading introduced by the interconnect, as quantified in Table II. In accordance with (1), the increased node capacitance raises the stage delay and therefore lowers the oscillation frequency relative to a

pre-layout estimate. This confirms that, at multi-gigahertz frequencies in a 65 nm process, parasitic-aware post-layout simulation is necessary to obtain a realistic figure; a schematic-only estimate would significantly overstate the achievable frequency.

Despite the compact  $73.96 \mu\text{m}^2$  core, the oscillator delivers a 6.50 GHz output while consuming only  $286.3 \mu\text{W}$ , demonstrating that the proposed methodology yields a verified, area- and power-efficient high-frequency oscillator suitable for integration in PLLs and on-chip clock generators. **[TODO: Optionally add a comparison table against prior art if your supervisor wants it; data for several references are available in [10].]**

## V. CONCLUSION

A three-stage CMOS ring oscillator was designed and realized through a complete, parasitic-aware physical-design methodology in a 65 nm CMOS technology. The flow—manual layout, functional DRC, LVS, and RC extraction—produced a verified layout from which a realistic post-layout characterization was obtained. The RC-extracted design oscillates at 6.50 GHz while consuming  $286.3 \mu\text{W}$  from a 1 V supply and occupying only  $73.96 \mu\text{m}^2$ . The results highlight the importance of parasitic-aware post-layout simulation for accurate high-frequency ring-oscillator design.

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